8 **(a)** A beam of electrons is directed along a normal towards a barrier, as shown in Fig. 8.1.

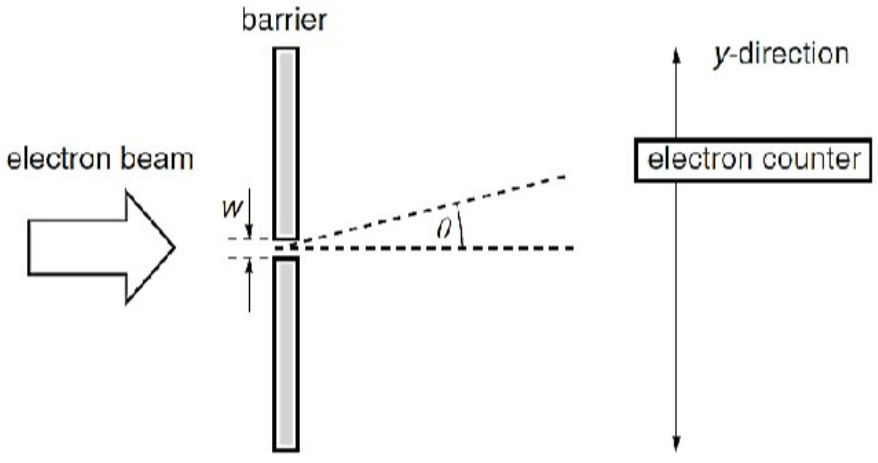


Fig. 8.1

The barrier contains a single slit of width w. Beyond the slit there is a counter that counts electrons. This can be moved in the y-direction to compare the rate of arrival of electrons at different values of the angle θ from the original direction of the beam.

(i) Louis de Broglie suggested that particles such as electrons have an associated wavelength. State the equation for the de Broglie wavelength of an electron and identify each term.

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..... [1]

(ii) State the condition for diffraction to be observable in this scenario.

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..... [1]

(iii) Electrons are accelerated by an accelerating voltage. State and explain what happens to angle θ when magnitude of the accelerating voltage decreases.

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..... [2]

(b) Werner Heisenberg used a different approach involving what is now known as the uncertainty principle. This can also be used to explain why the electron beam spreads out after passing through the slit. One version of this involves the equation

$$\Delta y \Delta p \gtrapprox h \, .$$

Using the uncertainty principle, explain what happens to angle θ when the slit is made narrower.

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..... [3]

(c) Explain whether your answer in **(b)** is consistent with single slit diffraction, when the slit is made narrower.

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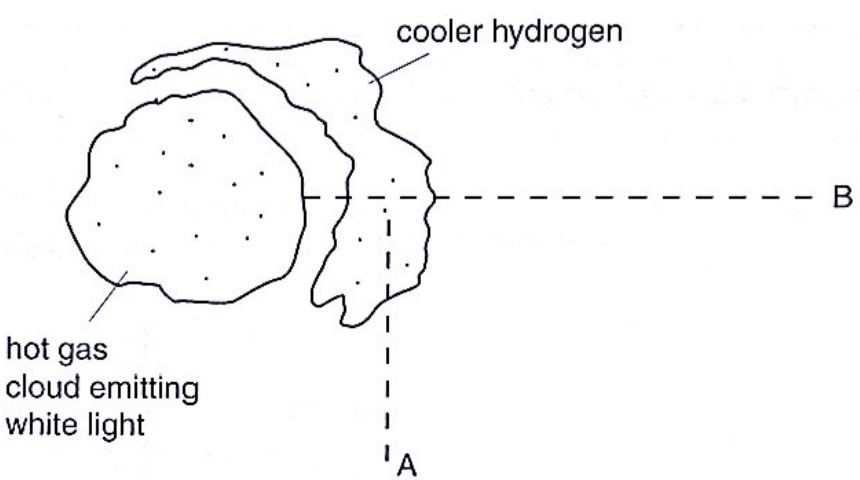
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..... [2]

(d) Fig. 8.2 shows a cooler region of hydrogen gas surrounding a hot gas cloud emitting white light.



(i) Describe the difference in appearance between an emission and absorption line spectra.

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.....

..... [1]

(ii) State the types of spectra seen from point A and B.

1. point A: .....

2. point B: ..... [1]

(iii) Explain how spectra seen at point B is obtained.

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..... [3]

(e) The energy levels E_n in a hydrogen atom are given by the empirical equation

$$E_n = \frac{-13.6\ \text{eV}}{n^2},$$

where n is an integer.

The four visible spectral lines P, Q, R and S in the hydrogen spectrum are illustrated in Fig. 8.3.

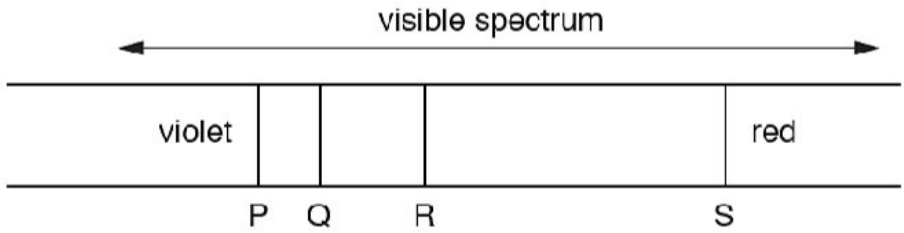


Fig. 8.3

The frequencies of three of these spectral lines are shown in the table.

spectral line	frequency / 10^{12} Hz
P	731
Q	f_Q
R	617
S	456

These frequencies are emitted when the energy level to which the electrons arrive is for $n = 2$.

(i) Deduce the value of n for the energy level from which an electron has moved in order to produce spectral line S.

n = ... [3]

(ii) Determine the frequency of the spectral line Q, f_Q.

f_Q = ... Hz [3]

[Total: 20]